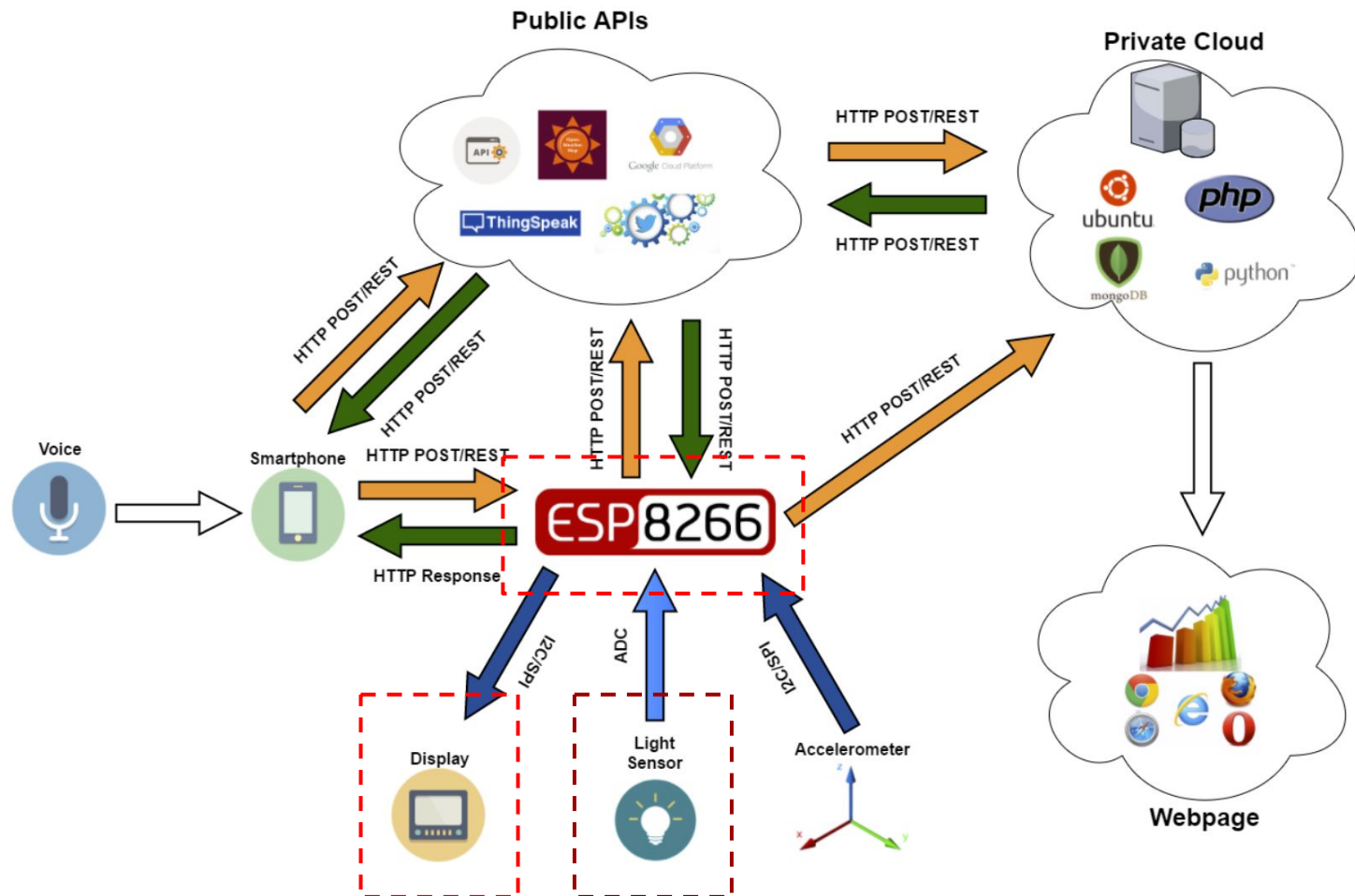
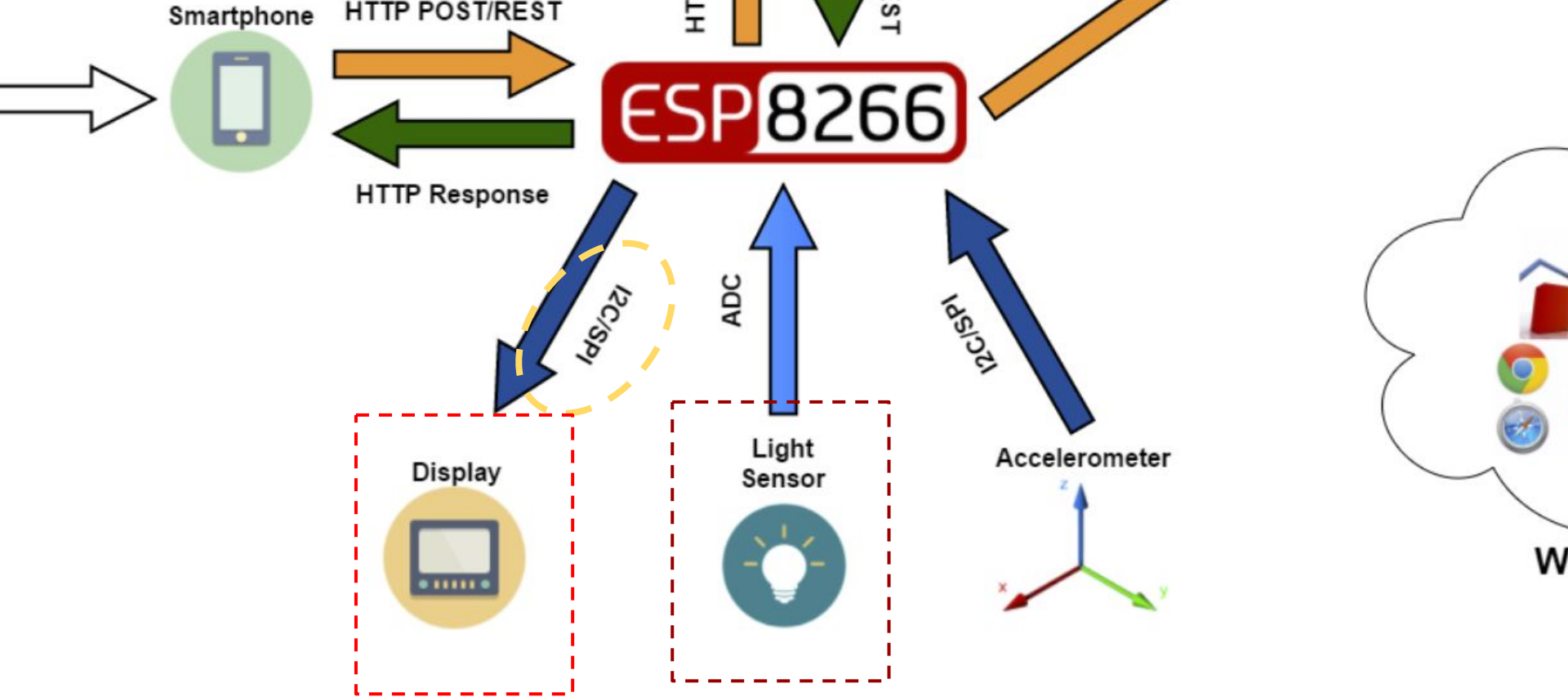


IoT - Intelligent and Connected Systems

EECS E4764 Fall' 22

Lab 3





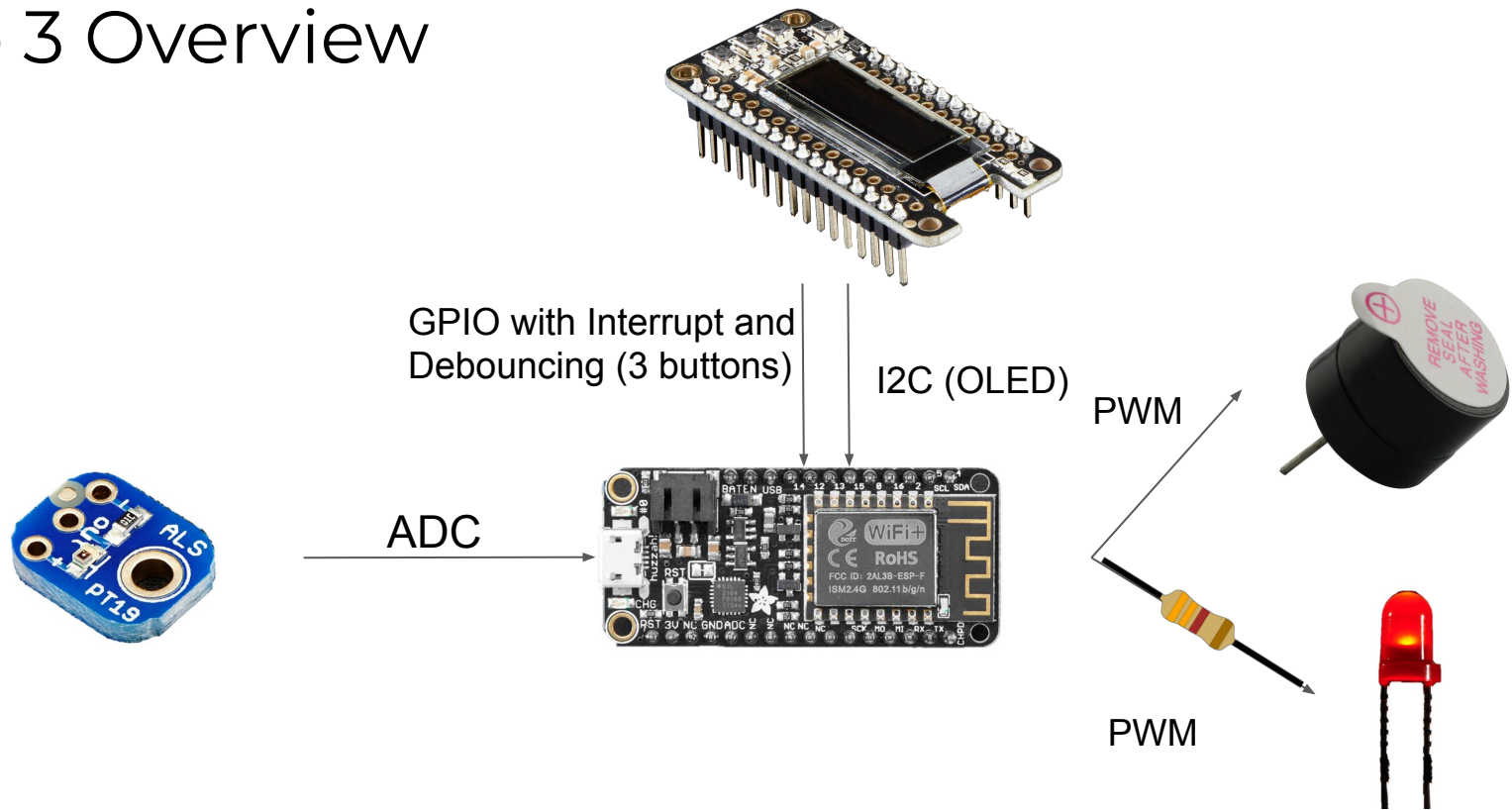
Lab 3: Apply I2C protocol to transmit data to/from display

Make your own interactive watch!

- Setup and update **time** properly like a normal watch
- Setup an **alarm** to go off at a certain time
- Make device **interactive** with **buttons**
 - **Debounce** buttons
- **Adjust screen brightness** based on ambient light (light sensor)



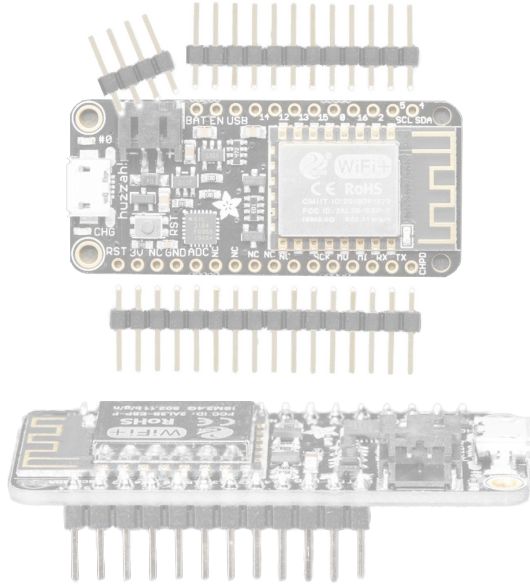
Lab 3 Overview



- <http://docs.micropython.org/en/v1.9.3/esp8266/esp8266/tutorial/pins.html>
- <https://learn.adafruit.com/adafruit-feather-huzzah-esp8266/pinouts>
- <https://docs.micropython.org/en/latest/esp8266/tutorial/ssd1306.html>

Soldering

Lab 2 (last week)

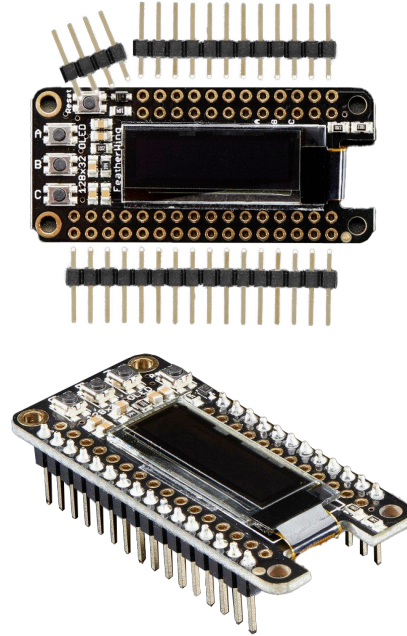


Feather HUZAZH

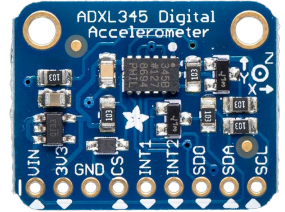


Light Sensor

Lab 3 (this week)



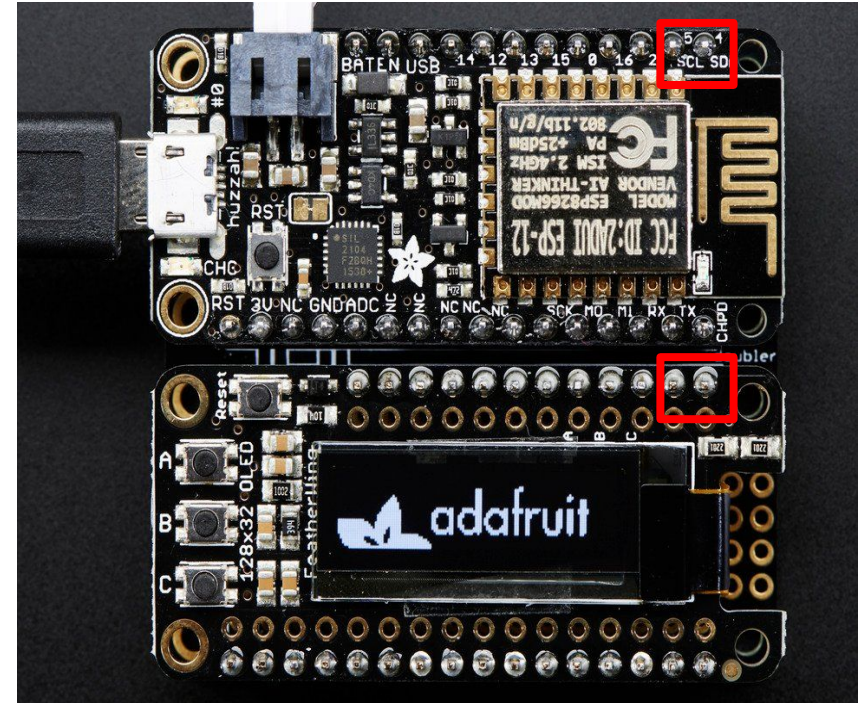
OLED



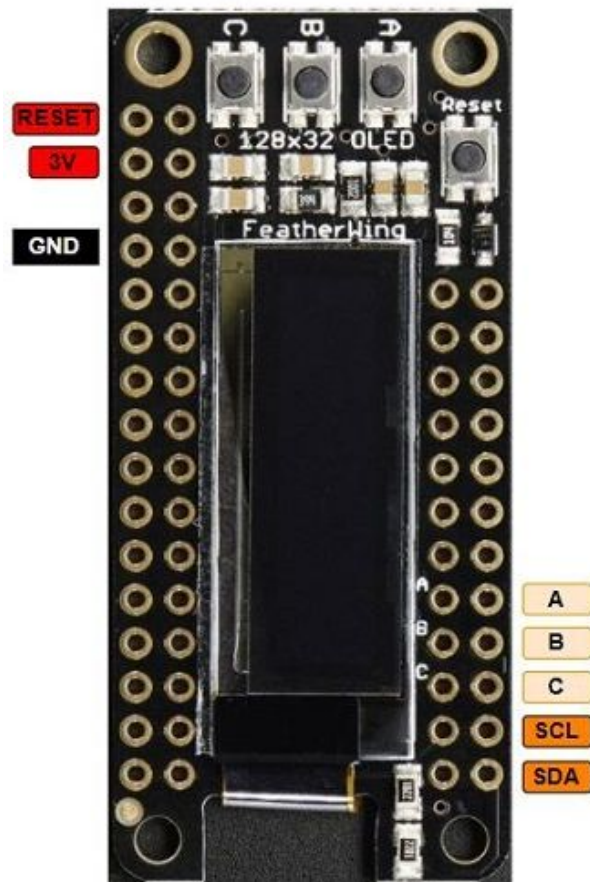
Accelerometer

Connecting OLED

- The Huzzah and OLED pins correspond one-to-one.
- GPIO Pins 4 & 5 = I2C pins for the connection.

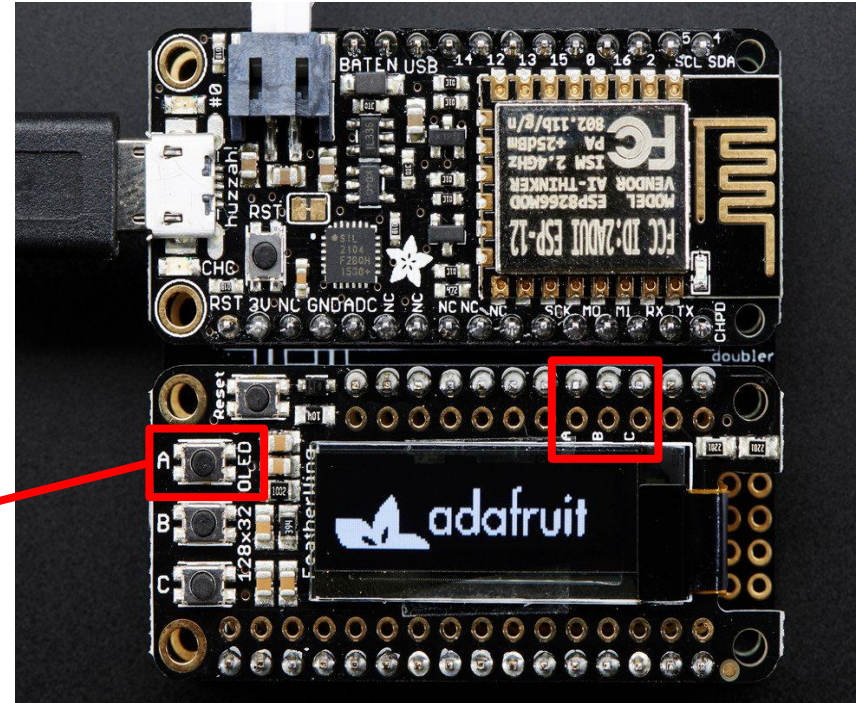
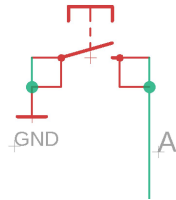


AdaFruit 128x32 OLED Display



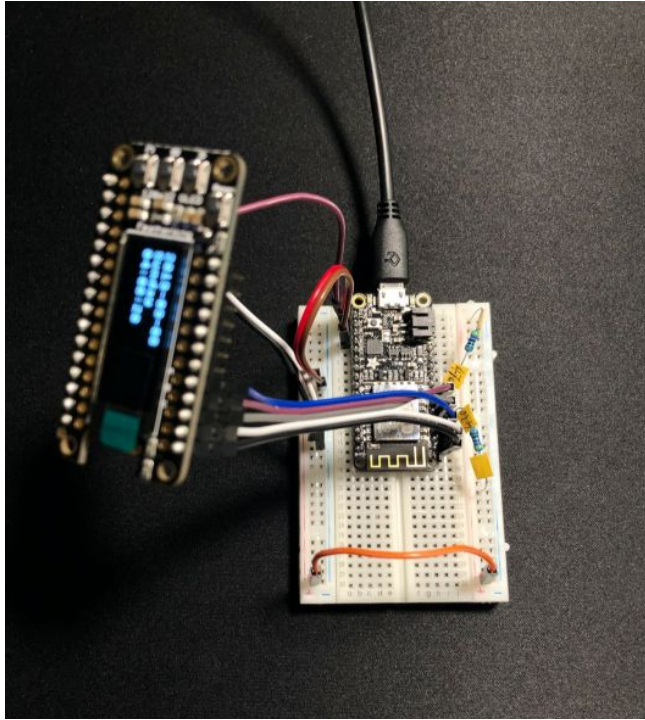
Connecting OLED

- Buttons A, B, C are connected to GND on the OLED PCB.
- Use any GPIO on HUZZAH that has **interrupt** and **internal pull-up** capability.



Lab 3 Checkpoints

1. Display the time on the OLED display; the time should update like a regular watch. Be able to change time through the OLED display buttons.
2. Adjust screen brightness based on ambient light.
3. Add alarm functionality; Be able to set the alarm through OLED display and when the alarm goes off, a visual and audio notification should go off.
4. Combine 1, 2, and 3.



Sample wiring

Soldering Queue Sign-up

Link also on Canvas

https://docs.google.com/spreadsheets/d/14TCjCHLViYTHt0h5gDicX2NHGen3E7xFYqZ_q1nHAnk/edit#gid=0